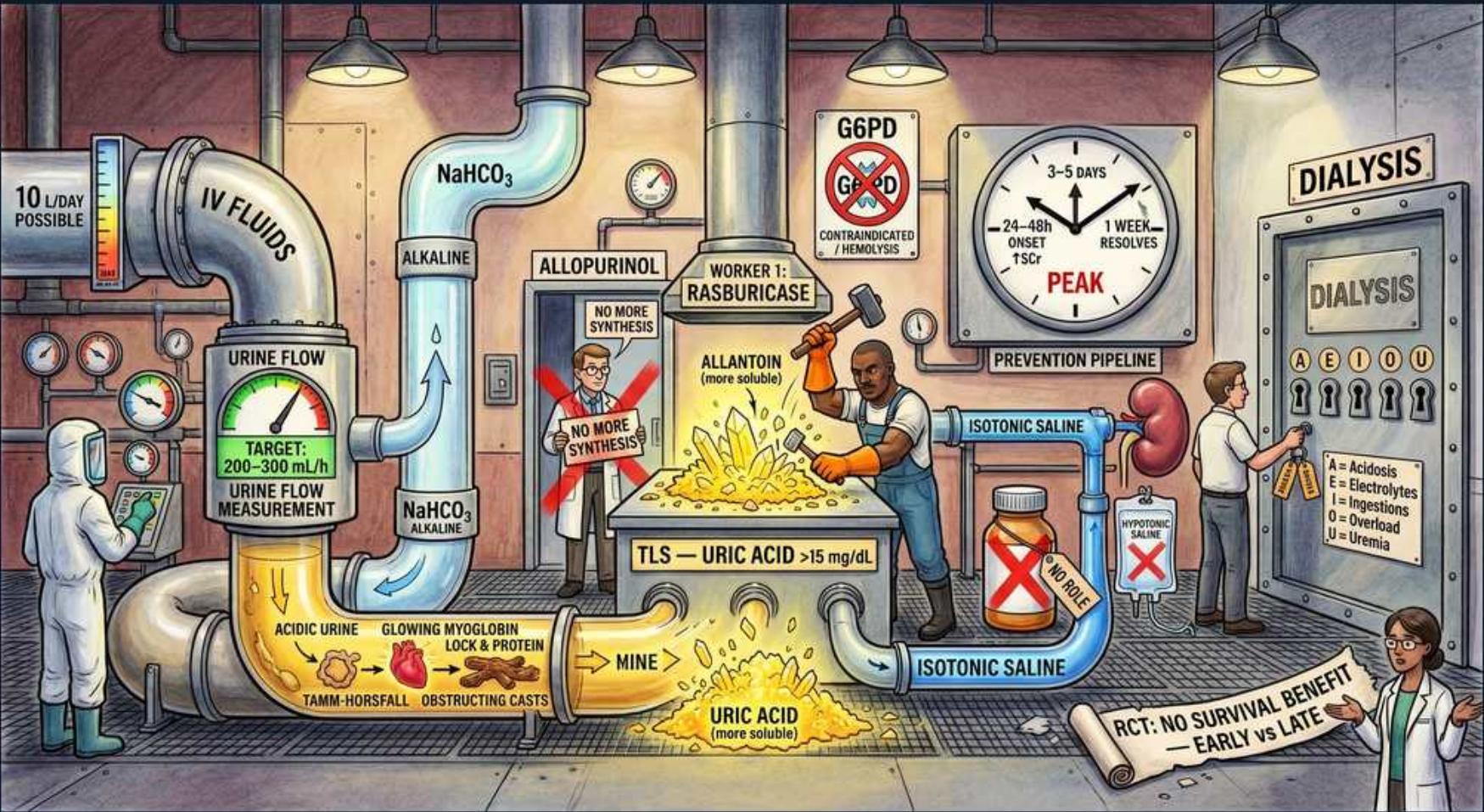


# Pigment Nephropathy & TLS: Prevention Pipeline





## 1. IV fluids — 10 L/day possible

### WHAT YOU SEE

Hazmat worker at 'IV FLUIDS, 10 L/day' tank

### WHAT IT ENCODES

Aggressive isotonic crystalloid is the cornerstone of rhabdomyolysis and TLS prevention. Volumes up to ~10 L/day flush obstructing casts and dilute nephrotoxins. Start before/early in the insult — once oliguric ATN is established fluids help far less.

### BOARD TRAP

Withholding fluids fearing overload — in rhabdo/TLS early aggressive volume is protective; the danger is under-resuscitation.

## 2. Target urine flow 200–300 mL/h

### WHAT YOU SEE

'URINE FLOW TARGET 200–300 mL/h' gauge

### WHAT IT ENCODES

Titrate IV fluids to a brisk urine output of 200–300 mL/h. High tubular flow prevents myoglobin/uric acid precipitation and cast obstruction. Monitor flow continuously rather than giving fixed volumes.

### BOARD TRAP

Aiming for normal UOP (>0.5 mL/kg/h) only — prophylaxis requires a supranormal 200–300 mL/h target.

## 3. Myoglobin lock & Tamm-Horsfall casts

### WHAT YOU SEE

'GLOWING MYOGLOBIN LOCK' + Tamm-Horsfall casts

### WHAT IT ENCODES

Myoglobin complexes with Tamm-Horsfall protein in acidic urine to form obstructing tubular casts; free iron also generates oxidative tubular injury and vasoconstriction. This is the mechanism of pigment-induced ATN.

### BOARD TRAP

Thinking myoglobin is filtered harmlessly — it is directly tubulotoxic and forms obstructing casts, especially in acidic urine.

## 4. Acidic urine worsens casts

### WHAT YOU SEE

'ACIDIC URINE' label on pipe

### WHAT IT ENCODES

Low urine pH promotes myoglobin cast formation and uric acid precipitation. This is the rationale offered for urinary alkalinization, though evidence beyond plain volume expansion is weak.

### BOARD TRAP

Over-prioritizing alkalinization — saline volume loading is the proven intervention; bicarb is adjunctive at best.

## 5. NaHCO<sub>3</sub> / alkaline urine

### WHAT YOU SEE

'NaHCO<sub>3</sub> ALKALINE' pipes

### WHAT IT ENCODES

Sodium bicarbonate raises urine pH to reduce cast formation (rhabdo) and uric acid crystallization (TLS). Caution: alkalinization can precipitate calcium phosphate in TLS and worsen hypocalcemia — not routinely superior to saline alone.

### BOARD TRAP

Routinely alkalinizing all TLS — it risks calcium-phosphate deposition; isotonic saline without bicarb is generally preferred.

## 6. Allopurinol blocks synthesis

### WHAT YOU SEE

'ALLOPURINOL — NO MORE SYNTHESIS' worker

### WHAT IT ENCODES

Allopurinol inhibits xanthine oxidase, PREVENTING new uric acid formation. Use prophylactically in low/intermediate-risk TLS. It does not lower already-formed uric acid and can cause xanthine nephropathy.

### BOARD TRAP

Expecting allopurinol to clear existing hyperuricemia — it only blocks synthesis; established high uric acid needs rasburicase.

## 7. Rasburicase converts to allantoin

### WHAT YOU SEE

'RASBURICASE → ALLANTOIN (more soluble)' worker

### WHAT IT ENCODES

Rasburicase (recombinant urate oxidase) enzymatically degrades existing uric acid to highly soluble allantoin, rapidly lowering levels. First-line for high-risk/established TLS with uric acid >8 mg/dL or AKI.

### BOARD TRAP

Using rasburicase in G6PD deficiency — it causes severe hemolysis (and methemoglobinemia) and is contraindicated.

## 8. G6PD contraindication

### WHAT YOU SEE

Crossed-out 'G6PD' hemolysis sign

### WHAT IT ENCODES

Rasburicase is contraindicated in G6PD deficiency because the H<sub>2</sub>O<sub>2</sub> byproduct triggers oxidative hemolysis and methemoglobinemia. Screen at-risk patients before administering.

### BOARD TRAP

Giving rasburicase empirically in high-risk ethnicities without G6PD screening — can precipitate life-threatening hemolysis.



## 9. TLS: uric acid >15, ↑K ↑PO4 ↓Ca

### WHAT YOU SEE

'TLS — URIC ACID >15 mg/dL' molten pile

### WHAT IT ENCODES

Tumor lysis releases intracellular contents: hyperuricemia, hyperkalemia, hyperphosphatemia, and secondary hypocalcemia. Calcium-phosphate and uric acid crystals obstruct tubules causing AKI. Highest risk in bulky high-turnover tumors (Burkitt, ALL).

### BOARD TRAP

Treating the hypocalcemia aggressively with calcium — giving calcium in hyperphosphatemic TLS risks metastatic calcification; treat only if symptomatic.

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## 10. Isotonic saline (not hypotonic)

### WHAT YOU SEE

'ISOTONIC SALINE' bottle, hypotonic crossed out

### WHAT IT ENCODES

Volume expansion uses isotonic saline to maintain high tubular flow and clear crystals. Hypotonic fluids are avoided. Diuretics/mannitol have NO proven role and may worsen obstruction.

### BOARD TRAP

Adding loop diuretics or mannitol to 'force diuresis' — no benefit, and they can aggravate cast obstruction and volume issues.

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## 11. 24–48h onset, peak 3–5 days, resolves ~1 wk

### WHAT YOU SEE

Clock '24–48h onset / 3–5d peak / 1 week resolves'

### WHAT IT ENCODES

Pigment/crystal AKI typically begins 24–48h after the insult, peaks at days 3–5, and recovers over ~1 week with supportive care if tubular integrity is preserved. Timeline mirrors ATN recovery phases.

### BOARD TRAP

Expecting immediate creatinine rise — the delay to 24–48h onset and 3–5 day peak is characteristic; early normal Cr does not exclude it.

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## 12. Dialysis indications — AEIOU

### WHAT YOU SEE

'DIALYSIS' vault with A-E-I-O-U

### WHAT IT ENCODES

Refractory cases need dialysis. AEIOU = Acidosis, Electrolytes (refractory hyperK), Ingestions/toxins, Overload (volume), Uremia. In TLS, dialysis clears phosphate/uric acid when medical therapy fails.

### BOARD TRAP

Delaying dialysis for an 'acceptable' creatinine — the indications are functional (refractory K, acidosis, overload, uremia), not an absolute Cr number.

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## 13. RCT: no survival benefit early vs late

### WHAT YOU SEE

Scroll 'RCT: no survival benefit — early vs late'

### WHAT IT ENCODES

Major trials (e.g., STARRT-AKI, AKIKI) show no survival benefit from early/prophylactic dialysis over a watchful strategy in AKI. Initiate dialysis for true indications rather than empirically.

### BOARD TRAP

Starting dialysis early 'to be safe' — evidence shows no mortality benefit; reserve it for AEIOU indications.

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